

Generation Structure and Coupling Matrix from $GF(27)$:

Trace Bilinear Form, Orbit Assignment, and CKM Complexity

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4 September 2026

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Abstract

The four irreducible polynomials $f_1, \dots, f_4$ over $\text{GF}(3)$ define the physical orbits in $\text{GF}(27)^*$. Their root orbits are genuine Frobenius orbits under $k \mapsto 3k \bmod 26$; the three elements correspond to the three generations. The trace bilinear form $\langle a, b \rangle = \text{Tr}_{\text{GF}(27)/\text{GF}(3)}(ab)$ gives the coupling structure: $f_3 \times f_4$ (charged leptons $\times$ quarks) is a permutation matrix **[B]**; $f_1 \times f_3$ (neutrinos $\times$ charged leptons) has normalised entries $1/\sqrt{2}$ **[B]**. Mixing angles (CKM, PMNS) themselves remain **[S]**.

.1 Why CKM mixing angles are complex

The CKM matrix V_{CKM} is unitary with four independent parameters (Wolfenstein parametrisation): $\lambda \approx 0.225$, $A \approx 0.811$, $\rho \approx 0.124$, $\eta \approx 0.356$. The CP-violating phase $\delta_{CP} \approx 68^\circ$ ($e^{i\delta_{CP}} = 0.375 + 0.927i$) makes the matrix essentially complex. It is responsible for CP violation in kaon and B-meson decays (observed 1964 and 2001 respectively).

What $\text{GF}(27)^*$ provides and what it does not

$\text{Gal}(\text{GF}(27)/\text{GF}(3)) = \mathbb{Z}_3$ has three irreducible representations over $\mathbb{C}$: trivial (1), $\omega = e^{2\pi i/3}$ (120°), ω^2 (240°). If the coupling matrix transforms under ρ_1 , a complex phase arises — the most plausible Galois candidate for the CP sector.

The problem: ω corresponds to 120° , not 68° (gap 52°) — no algebraically forced candidate.

Complete structure of the open problem **[S]**

Three steps are needed for the full CKM matrix from $\text{GF}(27)$: (1) **Real zero structure** — which couplings vanish: the trace bilinear form gives the permutation matrix (Theorem B) **[B]**. (2) **Mixing angles** — corrections to the permutation matrix giving $\lambda \approx 0.225$, $A \approx 0.811$: no forced candidate in $\text{GF}(27)$ **[S]**. (3) **CP phase** — a complex embedding of the Galois group forcing $\delta_{CP} \approx 68^\circ$: $\mathbb{Z}_3$ phases give 120° , not 68° **[S]**.

Core problem: $\text{GF}(27)^*$ is a finite field over $\text{GF}(3)$ with no natural complex structure. The trace bilinear form gives values in $\{0, 1, 2\} \subset \text{GF}(3)$. A complex embedding forcing $\delta_{CP} = 68^\circ$ does not currently exist **[S]**.

Important distinction: not derivable $\neq$ not usable. The absence of an algebraic derivation of δ_{CP} does not mean one cannot compute with it. The experimental value $\delta_{CP} \approx 68^\circ$ can be set as a phase anchor — structurally identical to $\nu = 246$ GeV or C_{conv} : empirically given, not freely adjustable, and once set all CP-violating observables (kaon, B-meson, neutrino CP) are fully determined. This is not a free parameter in the sense of a fit, but a phase anchor. Deriving δ_{CP} from first principles remains **[S]**; computing with it is immediately possible.

Bibliography

- [1] J. Pascher, Doc. 347, FFGFT corpus (Sept. 2026).
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